I. Internal portion:

1. Homographics 2. Wrapping images.

3. creating panaromoses 4. pinhole camera matrix.

5. post estimation from planes and Matrix

6. agument reality.

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1. Homographics.

* Homographics is a transformation used in computer vision to map points from one image plane to another image plane.

* It describe the relationship b/w two images of the same scene take from different view points.

Homography is represented using a 3x3 matrix that converts the coordinates of points from one image to another.

$$H = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}$$

if a point in the first image is (x, y) homography transform it to new point (x', y') in the second image.

Homographics :

$$\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$x' = Hx$

$y' = Hy$

Algorithms :

1. Direct Linear Transformations.
2. Affine Transformation.

Direct linear Transformation : (DLT)

The DLT algorithm is used to estimate the homography matrix b/w two images.

Affine Transformation :

Homography is used to map points from one image plane to another image plane. it finds the geometric relationship b/w 2 images takes from different view points.

Eq of Homeographics:

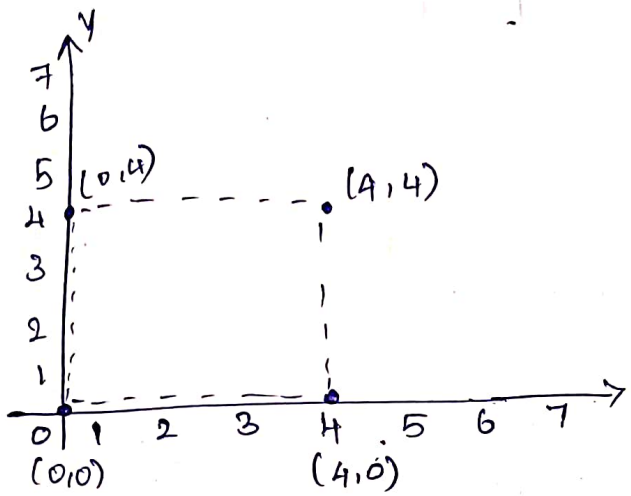
The Given point $\overset{x\ y}{(0,0)}$ $(4,0)$ $(4,4)$ $(0,4)$

$$H = \begin{bmatrix} 1.5 & 0 & 0 \\ 0 & 1.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

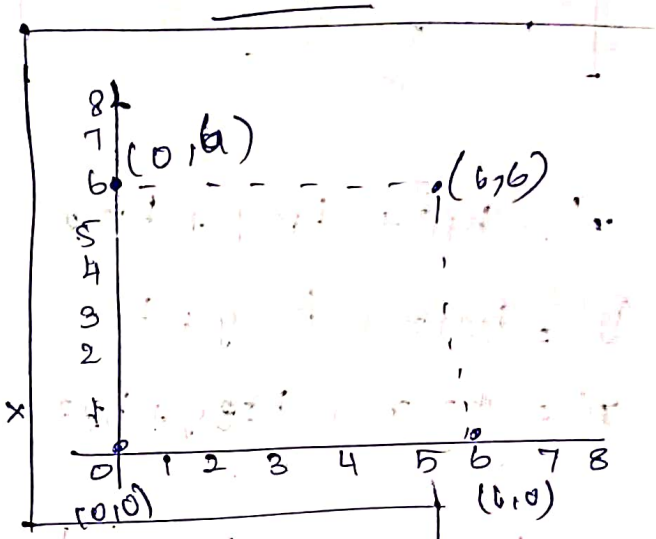
$1.5 \Rightarrow \dfrac{\text{New length}}{\text{old length}}$

$\dfrac{6}{4} = 1.5$

original point diagram.



1.5
4



$$\Rightarrow \begin{bmatrix} 1.5 & 0 & 0 \\ 0 & 1.5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

original point. New points.

(x, y)

$(0, 0) \Rightarrow x = 0 \times 1.5 = 0$, $y = 0 \times 1.5 = 0$ $\quad (x'y') = (0, 0)$

$(4, 0) \Rightarrow x = 4 \times 1.5 = 6$, $y = 0 \times 1.5 = 0$ $\quad (x'y') = (6, 0)$

$(4, 4) \Rightarrow x = 4 \times 1.5 = 6$, $y = 4 \times 1.5 = 6$ $\quad (x'y') = (6, 6)$

$(0, 4) \Rightarrow x = 0 \times 1.5 = 0$, $y = 4 \times 1.5 = 6$ $\quad (x'y') = (0, 6)$

Direct linear transformation:

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 1 \\ 1 & 3 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$$

$$x' = h_{11}x + h_{12}y + h_{13}$$

$$y' = h_{21}x + h_{22}y + h_{23}$$

$$z' = h_{31}x + h_{32}y + h_{33}$$

$$x' = \frac{h_{11}x + h_{12}y + h_{13}}{h_{31}x + h_{32}y + h_{33}}$$

$$x'[h_{31}x + h_{32}y + h_{33}] = h_{11}x + h_{12}y + h_{13}$$

$$x'x \quad x'y \quad x' \begin{bmatrix} h_{31} \\ h_{32} \\ h_{33} \end{bmatrix} - x - y - 1 \begin{bmatrix} h_{11} \\ h_{12} \\ h_{13} \end{bmatrix}$$

$$y' = \frac{h_{21}x + h_{22}y + h_{23}}{h_{31}x + h_{32}y + h_{33}}$$

$$y'[h_{31}x + h_{32}y + h_{33}] = [h_{21}x + h_{22}y + h_{23}]$$

$$y'x \quad y'y \quad y' \begin{bmatrix} h_{31} \\ h_{32} \\ h_{33} \end{bmatrix} - x - y - 1 \begin{bmatrix} h_{21} \\ h_{22} \\ h_{23} \end{bmatrix}$$

Derive transformation:

$$\begin{bmatrix} -x & -y & -1 & 0 & 0 & 0 & x'x & x'y & x' \\ 0 & 0 & 0 & -x & -y & -1 & y'x & y'y & y' \end{bmatrix} \begin{bmatrix} h_{11} \\ h_{12} \\ h_{13} \\ h_{21} \\ h_{22} \\ h_{23} \\ h_{31} \\ h_{32} \\ h_{33} \end{bmatrix}$$

(2.) pin hole camera Matrix.

Introduction:

It is also called the camera projection Matrix, which models how a 3D point in the world is projected onto a 2D image plane.

Basic Idea: A 3D point $x = (x, y, z)^T$

A 2D point $x = (x, y)^T$

Now represents these points in homogeneous coordinates

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \sim P \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

where P is the pin hole camera matrix.

The pinhole camera Matrix : $P = K\,[R/t]$ where K is the intrinsic calibration Matrix, R is the Rotation Matrix and t is the Translation vector.

Note : P is the 3 X 4 Matrix.

The intrinsic calibaration Matrix

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

where f_x, f_y is focal length in pixels and c_x, c_y is image center principal point.

The Extrinsic parameter $[R|t]$

$$[R|t] = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{bmatrix}$$

Example:

Let $f_x = 1000$, $f_y = 1000$, $c_x = 320$, $c_y = 240$

$R = I$ and $t = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ $\quad x = \begin{bmatrix} 1 \\ 2 \\ 5 \\ 1 \end{bmatrix}$

$$K = \begin{bmatrix} 1000 & 0 & 320 \\ 0 & 1000 & 240 \\ 0 & 0 & 1 \end{bmatrix} \qquad [R|t] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$P = K[R|t]$

$$= \begin{bmatrix} 1000 & 0 & 320 \\ 0 & 1000 & 240 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$P = \begin{bmatrix} 1000 & 0 & 320 & 0 \\ 0 & 1000 & 240 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$x = PX$

$$\begin{bmatrix} 1000 & 0 & 320 & 0 \\ 0 & 1000 & 240 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 5 \\ 1 \end{bmatrix}$$

$(1000 \times 2) + (0 \times 2) + (320 \times 5) + (0 \times 1)$

$= 2000 + 1600$

$= 3600.$

$(1000 \times 1) + (0 \times 2) + (320 \times 5) + (0 \times 1)$

$= 1000 + 0 + 1600 + 0$

$= 2600.$

$$X = \begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} 2600 \\ 3200 \\ 5 \end{bmatrix}$$

$x = x'/w' = 2600/5 = 520$

$y = y'/w' = 3200/5 = 640.$

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(3) Pole Estimation from planes and semaries.

Meaning of pole estimation:

* pole estimation means finding the position of an object (like a pole or vertical line) in an image or 3D scene.

* It is used to estimate the location or direction of a vertical object in an image.

(1) Planes in pole estimations:

* A plan is a flat 2D surface in 3D space.

* It is used to understand scene geometry, perform homograpy transformation, image mapping,

Steps:

1. Word point $P = (X, Y, Z, 1)$ assume $Z = 0$

$$P = (X, Y, 0, 1)$$

2. projecting image $P = K[R|t]P$

$$P = K[r_1 \; r_2 | t] \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix} \quad r_3 = 0$$

$\therefore H = K[r_1 \; r_2 | t]$ Where K is a camera Intrinsic Matrix r_1 and r_2 is 1st & 2nd column of rotation matrix and t is translation vector.

3. Entracking pole from homography:

$K^{-1} H = [h_1, h_2, h_3]$, $r_1 = \lambda h_1$ $r_2 = \lambda h_2$ and

$t = \lambda h_3$ where $\lambda = \frac{1}{\|h_1\|}$

$r_3 = r_1 \times r_2$ using cross product

$$R = [r_1 \; r_2 \; r_3]$$

(2) Squares in pole estimation.

A Square is a four-sided shape where all sides are equal and all angles are 90°

In computer Vision squares are often used for pattern detection, marker ~~det~~ tracking, camera calibration.

eg: A QR code or chessboard pattern contains squares.

Steps:

1. Detect the 4 corners of the squares in the image
2. Assign known 3D coordinates to the square corners
3. To find R and t use perspective point algorithm.

Ex: $P_1 = (0,0,0)$ $P_2 = (1,0,0)$ $P_3 = (1,1,0)$ $P_4 = (0,1,0)$

$P_1 = (100, 100)$ $P_2 = (200, 100)$, $P_3 (200, 200)$ $P_4 = (100, 200)$

$$K = \begin{bmatrix} 100 & 0 & 0 \\ 0 & 100 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad H = \begin{bmatrix} 100 & 0 & 100 \\ 0 & 100 & 100 \\ 0 & 0 & 1 \end{bmatrix}$$

diagonal $(1/a)$

$$K^{-1}H = \begin{bmatrix} 0.01 & 0 & 0 \\ 0 & 0.01 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 100 & 0 & 100 \\ 0 & 100 & 100 \\ 0 & 0 & 1 \end{bmatrix}$$

1st row

$(0.01 \times 100) + (0 \times 0) + (0 \times 0) = 1$

$\cancel{(0 \times 0) + (0.01 \times 100) + (0 \times 100)} = 0$

$(0.01 \times 0) + (0 \times 0) + (0 \times 100) = 0$

$(0.01 \times 100) + (0 \times 0) + (0 \times 100) = 1$

$$K^{-1}H = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$r_1 = (1,0,0)$ $r_2 = (0,1,0)$ $t = (1,1,1)$

$$r_3 = r_1 \times r_2 \Rightarrow \begin{bmatrix} i & j & k \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\begin{vmatrix} 0 & 0 \\ 1 & 0 \end{vmatrix} i - \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} j + \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$$

$0 - 0 + 1 \Rightarrow (0,0,1)$

$$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\therefore R = I$, and $t = (1, 1, 1)$

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(4) Augamented Reality:

Definition:

Augument Reality is technology that add virtual objects such as images, 3D models or graphics on the top of the real world enviorment in real time using camera or display device.

It combines the real world + computer generated content.

eg: Snapchat filters, AR Games, mobile camera effects.

Working principle of AR:

1. Captured Real scene: The camera ~~capture~~ captures the Real world environment.

2. Feature detection:

Important points or features of the scence are detected using algorthims such as SIFT or Harris corner ~~Dect~~ Detector.

3. camera pose Estimation: The system calculates the position and orientation of the camera.

4. 3D object projection:

A virtual 3D-object is projected onto the real image using the camera matrix.

5. Rendering:

Finally the virtual object is displayed on the real world scene with proper alignment.

A 3D ~~image~~ object projected onto image using the camera matrix $P = K[R|t]P$ where P is 2D image point P is a 3D virtual object point, K is a camera intrinsic matrix, R is a Rotation matrix and t is translation vector.

Eg:

$$K = \begin{bmatrix} 1000 & 0 & 320 \\ 0 & 1000 & 240 \\ 0 & 0 & 1 \end{bmatrix} \quad R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$t = \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} \quad P = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

$$P = K[R|t]\,P$$

$$[R|t] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 5 \end{bmatrix}$$

$$[R|t]\,P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix}$$

$$P = K[R|t]P = \begin{bmatrix} 1000 & 0 & 320 \\ 0 & 1000 & 240 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix} = \begin{bmatrix} 2600 \\ 2200 \\ 5 \end{bmatrix}$$

$$P(x, y) = [2600/5,\ 2200/5] = [520,\ 440]$$

Tools are used AR:

1. from open GL . GL Import *
2. from open GL . GLU Import *
3. import pygame . pygame . image .
4. from pygame . local import * .

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⑤ Wrapping Images:

Introduction:

Image wraping is a process used to transform an image from one coordinate system to another. It changes the positions, shapes or perpective of an image using mathematical tranformation.

Image wrapping widely used.

Image stitching

Image registration

* Augmented reality
* computer graphics.

1. Image in Image:

* Image - in Image place one image inside another image using geometric transformation.

* The Image is mapped to a specific region of the second image

eg: A moive poster placed on billboard in a street photo.

Steps: 1. select the region in the target image.
2. capture transformation matrix.
3. wrap the source image into the region.

2. piece wise wrapping:

* piece wise wrapping divides an image into small regions (triangles or squares) and transforms each region separately.

* This method gives more accurate transformation when images are complex

eg: Face morphing application.

Steps: 1. Divide image into triangles
2. Apply transformation to each triangle
3. combine them to form the final wrapped image.

4. Registering images:

Image Registration is the process of aligning 2 or more Images of the same scene. It finds the correct transformation so that the images match prefectly.

eg: satellite images taken at different image are aligned to detect changes in land or buildings.

Application:

Medical imaging

Remote sensing

panorama creation.